Investigation: AMERICANAS S.A. (CVM code 20990), fiscal year 2020, as it was on 2022-04-30

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Comércio (Atacado e Varejo) (origin: FCA); group: non-financial.
Trading codes (2021 FCA, version 3, filed on 2022-03-29): BTOW3 (Ações Ordinárias, Bolsa, since 2007-08-08).

Document used

2020 DFP, version 2, filed on 2021-03-04; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2021-03-04); 2 (2021-03-04).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=100690

Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: −1.550 (index)
- Alert by the study 1 rule: yes
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: 79 (n = 40)
- Percentile in group and year: 86 (n = 470)
- The company's own history: median −2.515; Tukey range (1.5 × IQR): from −3.809 to −1.692; value above the range (index; 5 fiscal years)
- Confidence in the data: medium

DSRI, days' sales in receivables index
- Value: 1.613 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 78 (n = 44)
- Percentile in group and year: 85 (n = 576)
- The company's own history: median 0.841; Tukey range (1.5 × IQR): from 0.550 to 1.022; value above the range (index; 5 fiscal years)
- Confidence in the data: medium

GMI, gross margin index
- Value: 1.046 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 63 (n = 45)
- Percentile in group and year: 70 (n = 594)
- The company's own history: median 0.905; Tukey range (1.5 × IQR): from 0.677 to 1.188; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium

AQI, asset quality index
- Value: 0.737 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 17 (n = 45)
- Percentile in group and year: 10 (n = 586)
- The company's own history: median 1.010; Tukey range (1.5 × IQR): from 0.654 to 1.268; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium

SGI, sales growth index
Value: 1.496 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 88 (n = 45)
Percentile in group and year: 89 (n = 597)
The company's own history: median 1.032; Tukey range (1.5 × IQR): from 0.821 to 1.176; value above the range (index; 5 fiscal years)
Confidence in the data: medium

DEPI, depreciation index
Value: 0.911 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 48 (n = 41)
Percentile in group and year: 33 (n = 494)
The company's own history: median 0.844; Tukey range (1.5 × IQR): from 0.813 to 0.867; value above the range (index; 5 fiscal years)
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 0.912 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 26 (n = 45)
Percentile in group and year: 32 (n = 585)
The company's own history: median 1.078; Tukey range (1.5 × IQR): from 0.672 to 1.475; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.007 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 68 (n = 46)
Percentile in group and year: 70 (n = 616)
The company's own history: median −0.016; Tukey range (1.5 × IQR): from −0.078 to 0.051; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

LVGI, leverage index
Value: 0.903 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 28 (n = 45)
Percentile in group and year: 19 (n = 604)
The company's own history: median 0.991; Tukey range (1.5 × IQR): from 0.796 to 1.249; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

DPO jump
Value: −6.8 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 52.7 days)
Percentile in sector and year: 21 (n = 45)
Percentile in group and year: 21 (n = 570)
The company's own history: median 4.4 days; Tukey range (1.5 × IQR): from −31.7 days to 54.2 days; value inside the range (5 fiscal years)
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix

(decision of 2026-11-07)

Implied interest rate minus average CDI
Value: 8.20 p.p.
Alert by the study 1 rule: no
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 27 (n = 42)
Percentile in group and year: 41 (n = 560)
The company's own history: median 16.00 p.p.; Tukey range (1.5 × IQR): from −7.99 p.p. to 41.69 p.p.; value inside the range (5 fiscal years)
Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.008 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 68 (n = 46)
Percentile in group and year: 70 (n = 616)
The company's own history: median −0.018; Tukey range (1.5 × IQR): from −0.086 to 0.056; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

Financial risk (context only)

Altman Z" (financial risk, context only)
Value: 3.440 (index)
Components: X1 = 0.462; X2 = −0.124; X3 = 0.008; X4 = 0.723 (indices)
Percentile in sector and year: 75 (n = 46)
The company's own history: median 1.740; Tukey range (1.5 × IQR): from 1.454 to 2.037; value above the range (index; 5 fiscal years)
- No cut-off zone: the available source does not define zones for Z"
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 4/9 (8 computable)
Signals:
- Return on assets (ROA): 0
- Operating cash flow on assets (CFO): 0
- Change in ROA (ΔROA): 1
- Cash above earnings (ACCRUAL): 1
- Change in leverage (ΔLEVER): 0
- Change in current ratio (ΔLIQUID): 1
- No equity issuance (EQ_OFFER): —
- Change in gross margin (ΔMARGIN): 0
- Change in asset turnover (ΔTURN): 1
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

Changes

Highlighted in the following year's comparative: none.
Highlighted: differences in totals and key lines (total assets, current liabilities, non-current liabilities, equity, net revenue, net income and operating cash flow) above 1% of total assets of the version used. The other differences (reclassifications between lines and smaller differences) stay in the full list.
Full list: 13 accounts above the R$1,000 tolerance (group a, totals and key lines: 2; group b, reclassifications: 11), in the annex at the end.

What was known on the date

- DFP 2015, v1: 2016-03-10
- DFP 2015, v2: 2016-03-15
- DFP 2016, v1: 2017-02-20
- DFP 2017, v1: 2018-03-07
- DFP 2018, v1: 2019-03-20
- DFP 2019, v1: 2020-02-20
- DFP 2019, v2: 2020-02-21
- FCA 2020, v1: 2020-05-29
- DFP 2020, v1: 2021-03-04
- DFP 2020, v2: 2021-03-04
- DFP 2021, v1: 2022-02-24

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

Data quality

53 applicable checks; 0 failed.

Auditor's report and auditor (factual section; no alert)

Auditor's report: Sem Ressalva (unqualified).
no opinion excerpt was found in the text
Auditor in the 2019 FCA: PricewaterhouseCoopers Auditores Independentes.
Auditor in the 2020 FCA: PricewaterhouseCoopers Auditores Independentes.
Auditor in the 2021 FCA: PricewaterhouseCoopers Auditores Independentes.
Auditor changes (one line per change, canonical name by the firm's CVM code):

- 2011: Ernst & Young Auditores Independentes S.S. -> PricewaterhouseCoopers Auditores Independentes (transition: two firms in the 2011 FCA). The outgoing firm audited 2 consecutive fiscal years (2009 to 2010), by the start and end dates in the FCA.
- 2016: PricewaterhouseCoopers Auditores Independentes -> KPMG Auditores Independentes. The outgoing firm audited 5 consecutive fiscal years (2011 to 2015), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).
- 2019: KPMG Auditores Independentes -> PricewaterhouseCoopers Auditores Independentes. The outgoing firm audited 3 consecutive fiscal years (2016 to 2018), by the start and end dates in the FCA.

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-

years (docs/summary.md).

Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 38 DFP lines, each with document, version, filing date, account and location.

Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E). Generated at 2026-11-10T19:37:22+00:00.

Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.

Annex: full list of differences in the following year's comparative (R$ million)

(a) BPP 2.01 Passivo Circulante: 5,331.1 in the version used, 5,386.8 in the comparative (difference of 55.8)
(a) BPP 2.02 Passivo Não Circulante: 7,790.0 in the version used, 7,734.2 in the comparative (difference of −55.8)
(b) BPP 2.01.06 Provisões: 0.0 in the version used, 55.8 in the comparative (difference of 55.8)
(b) BPP 2.01.06.01 Provisões Fiscais Previdenciárias Trabalhistas e Cíveis: 0.0 in the version used, 55.8 in the comparative (difference of 55.8)
(b) BPP 2.02.04 Provisões: 161.2 in the version used, 105.5 in the comparative (difference of −55.8)
(b) BPP 2.02.04.01 Provisões Fiscais Previdenciárias Trabalhistas e Cíveis: 161.2 in the version used, 105.5 in the comparative (difference of −55.8)
(b) BPP 2.03.02.07 Plano de Opção de Ações: 47.1 in the version used, 0.0 in the comparative (difference of −47.1)
(b) BPP 2.01.06.01.02 Provisões Previdenciárias e Trabalhistas: 0.0 in the version used, 38.9 in the comparative (difference of 38.9)
(b) BPP 2.02.04.01.02 Provisões Previdenciárias e Trabalhistas: 84.9 in the version used, 46.0 in the comparative (difference of −38.9)
(b) BPP 2.01.06.01.04 Provisões Cíveis: 0.0 in the version used, 16.9 in the comparative (difference of 16.9)
(b) BPP 2.02.04.01.04 Provisões Cíveis: 43.1 in the version used, 26.2 in the comparative (difference of −16.9)
(b) DFC (indirect method) 6.01.02 Variações nos Ativos e Passivos: −713.1 in the version used, −718.7 in the comparative (difference of −5.6)
(b) DFC (indirect method) 6.01.03 Outros: −5.6 in the version used, 0.0 in the comparative (difference of 5.6)